Shiv Nadar University Chennai

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

Experiment 1
Implementation of a Single Layer Perceptron for Binary Classification

1. Objective

The objective of this experiment is to understand the concept of an artificial neuron and implement a Single Layer Perceptron from scratch for binary classification. Students will learn the perceptron learning algorithm, understand the importance of activation functions, visualize the learning process, and evaluate the classifier using a real-world dataset.

2. Background Theory

An Artificial Neuron is the fundamental building block of a neural network. It receives one or more input features, multiplies them by corresponding weights, adds a bias term, and applies an activation function to determine the output.

The Single Layer Perceptron is the earliest supervised learning algorithm capable of solving linearly separable binary classification problems.

Mathematical Formulation

Weighted Sum

$$z = \mathbf{w}^T \mathbf{x} + b$$

where

- $\mathbf{x}$: Input vector
- $\mathbf{w}$: Weight vector
- b : Bias
- z : Linear combination

Prediction

$$\hat{y} = f(z)$$

Weight Update Rule

$$\mathbf{w}_{new} = \mathbf{w}_{old} + \eta(y - \hat{y})\mathbf{x}$$

Bias Update

$$b_{new} = b_{old} + \eta(y - \hat{y})$$

where η denotes the learning rate.

3. Activation Functions

An activation function determines the output of a neuron based on the weighted sum of its inputs. It introduces non-linearity, enabling neural networks to learn complex patterns. Although the perceptron uses only the Step Activation Function, modern deep learning models employ differentiable activation functions for efficient optimization through backpropagation.

Step Activation Function

$$f(z) = \begin{cases} 1, & z \geq 0 \\ 0, & z < 0 \end{cases}$$

The Step Activation Function produces a binary output and is therefore suitable only for linearly separable binary classification problems.

Activation	**Output Range**	**Typical Applications**
Step	$\{0, 1\}$	Classical Perceptron
Sigmoid	$(0, 1)$	Binary Classification Output Layer
Tanh	$(-1, 1)$	Hidden Layers (Traditional Networks)
ReLU	$[0, \infty)$	Hidden Layers in CNNs and MLPs
Leaky ReLU	$(-\infty, \infty)$	Avoiding Dying ReLU
Softmax	$(0, 1)$	Multi-class Classification Output Layer

Note: This experiment uses only the Step Activation Function. The remaining activation functions will be studied in subsequent experiments involving Multilayer Perceptrons and Deep Neural Networks.

4. Dataset Description

Dataset: Banknote Authentication Dataset
Source: UCI Machine Learning Repository
Download Link:
`https://archive.ics.uci.edu/dataset/267/banknote+authentication`

Instances	1372
Features	4 Numerical Features
Classes	2
Missing Values	None
Task	Binary Classification

Feature Description

- Variance
- Skewness
- Curtosis
- Entropy

Target Class

- 0 – Authentic Banknote
- 1 – Forged Banknote

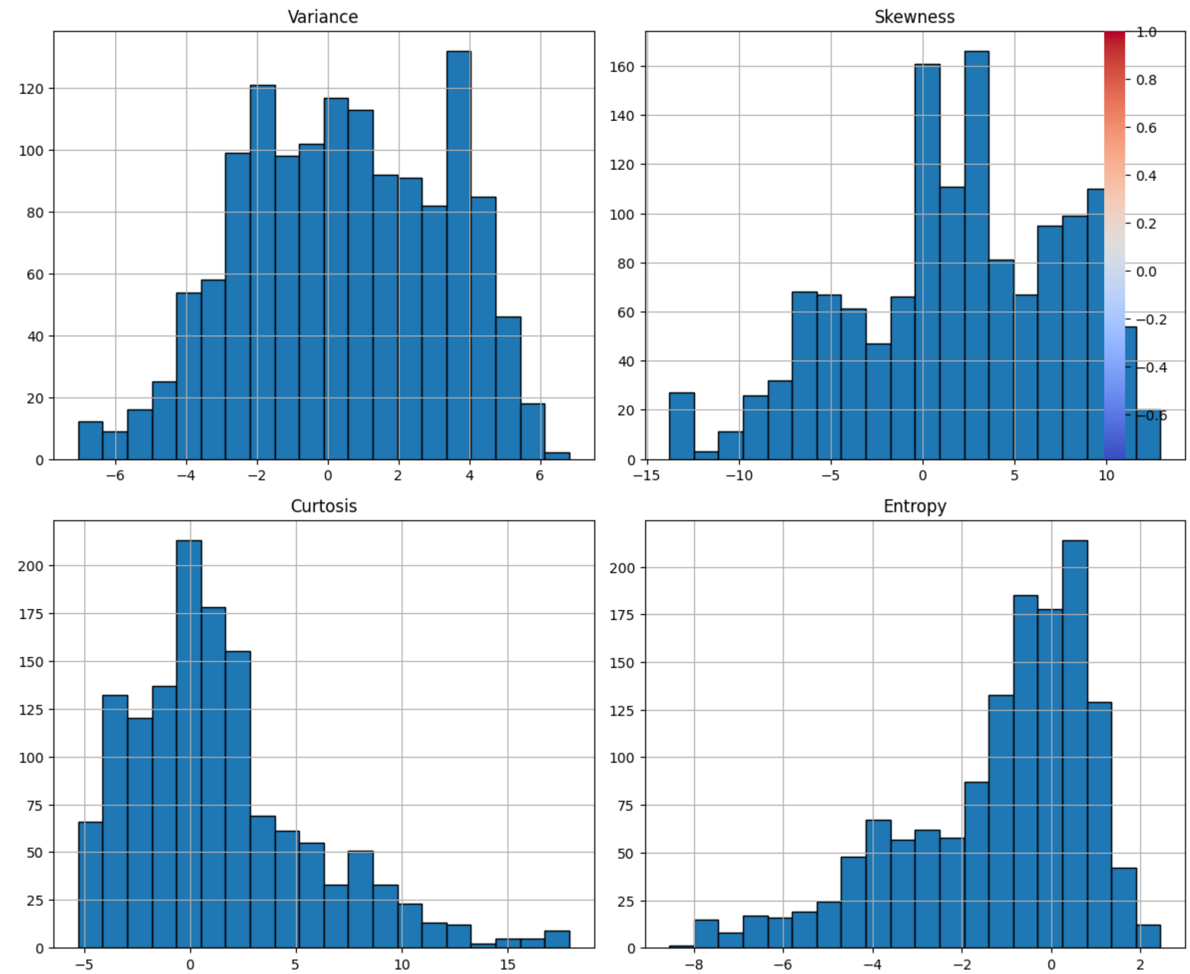
5. Source Code

The complete source code for this experiment, including dataset loading, exploratory data analysis, the from-scratch Single Layer Perceptron implementation, training, evaluation, and all plot generation, is available at the following GitHub repository:

`https://github.com/UvanAdhithya/DeepLearning-Lab`

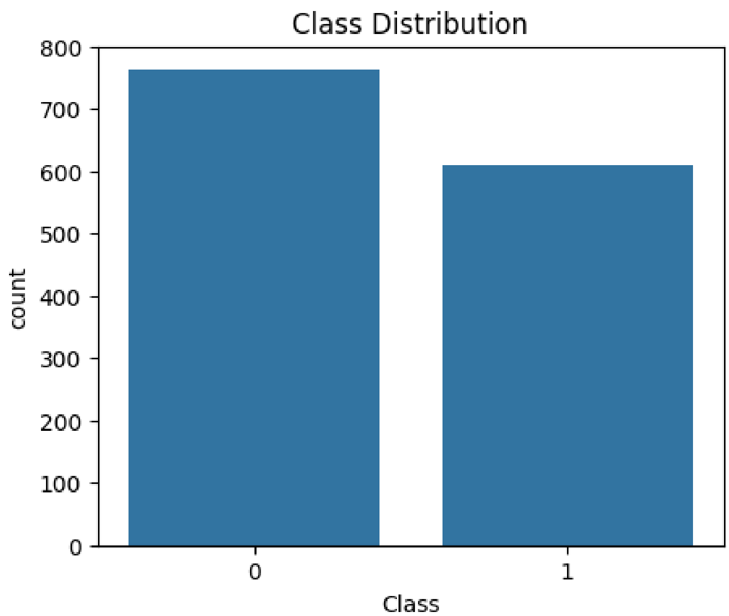
6. Mandatory Plots

Feature Histograms, Correlation Heatmap, Scatter Plot and Boxplots



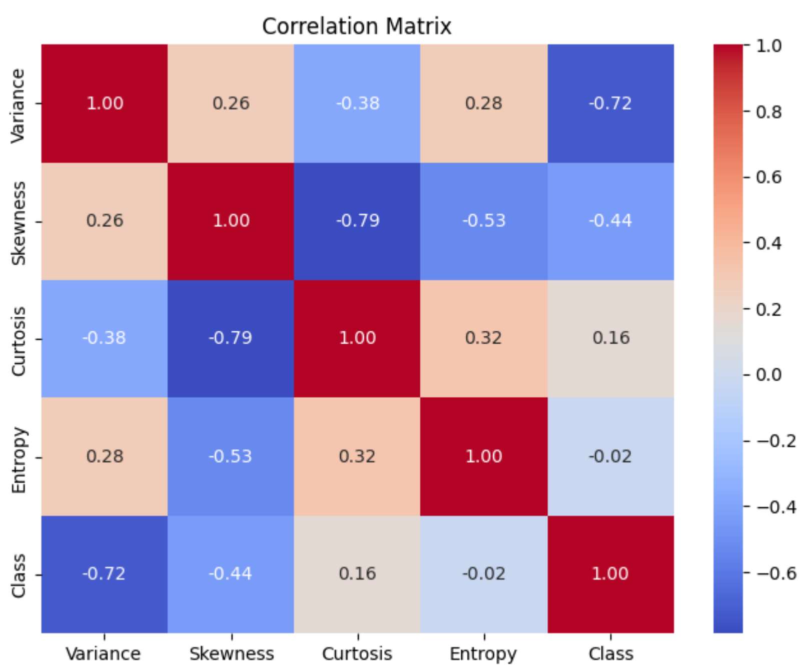
The representation in histograms reveals that the distribution of the Variance and Entropy follows a roughly symmetric distribution whereas under the criteria of Skewness and Curtosis, it is noted as highly right skewed while the most serious outlier in the higher part of this distribution follows the criteria of Curtosis. The correlation heatmap shows that there is a strong inverse correlation of Class with Variance and moderate inverse correlation between Curtosis and Skewness, while the scatterplot of Variance and Skewness shows that the two classes are largely but not perfectly linearly separable in this regard. The boxplot confirms that in these four features the parameters Skewness and Curtosis follow the criteria of having the largest spread and number of outliers.

Class Distribution



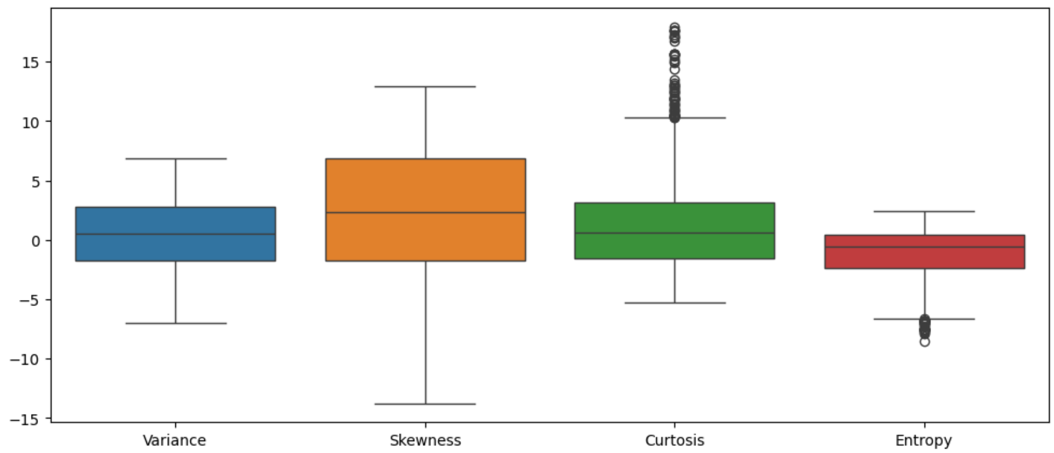
The distribution of classes indicates approximately 762 authentic notes of class 0 and 610 forged notes of class 1 which makes the data set mildly imbalanced. This kind of near-balance allows for considering accuracy as a reliable metric of evaluation, which means there is no need for introducing techniques like class weighting or resampling.

Correlation Matrix



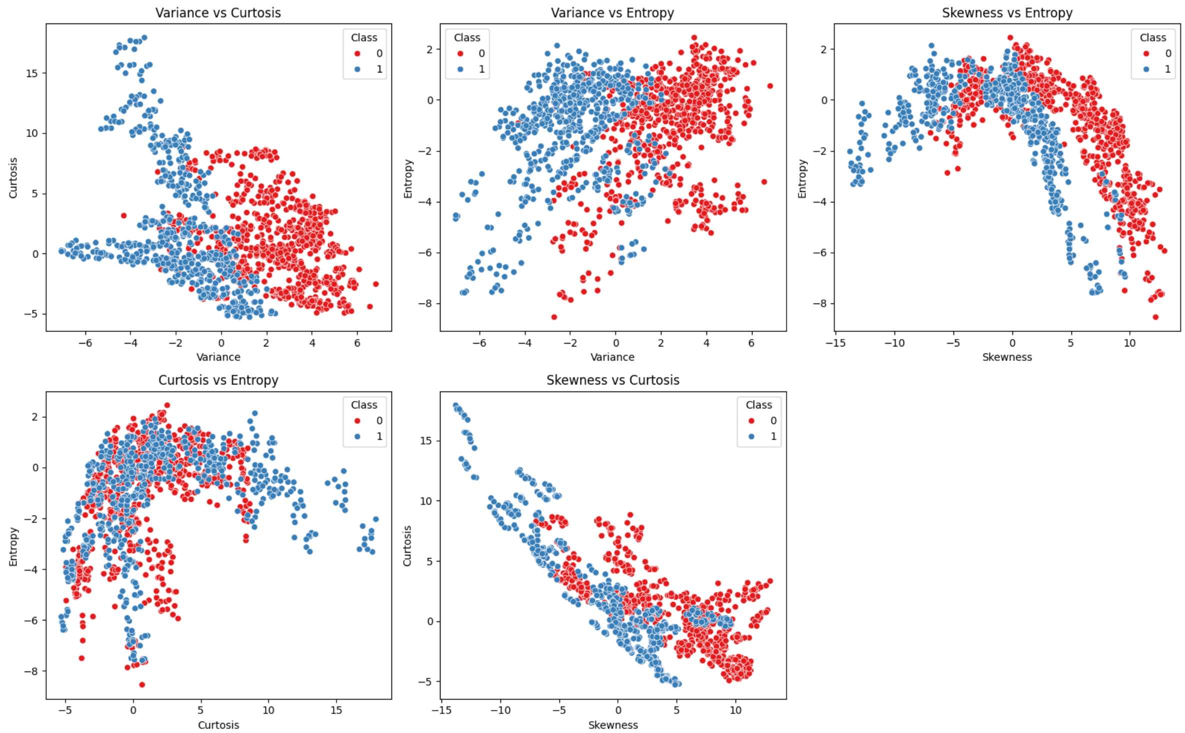
Variance has the strongest relationship with the target variable, confirming it is the most discriminative feature. Skewness is also moderately correlated with the target variable. Moreover, given the strong negative coefficient of correlation between Skewness and Curtosis, one may assume that one of these variables is redundant. The Entropy variable is weakly correlated with the target Class, contributing to the least discrimination.

Boxplots of Features



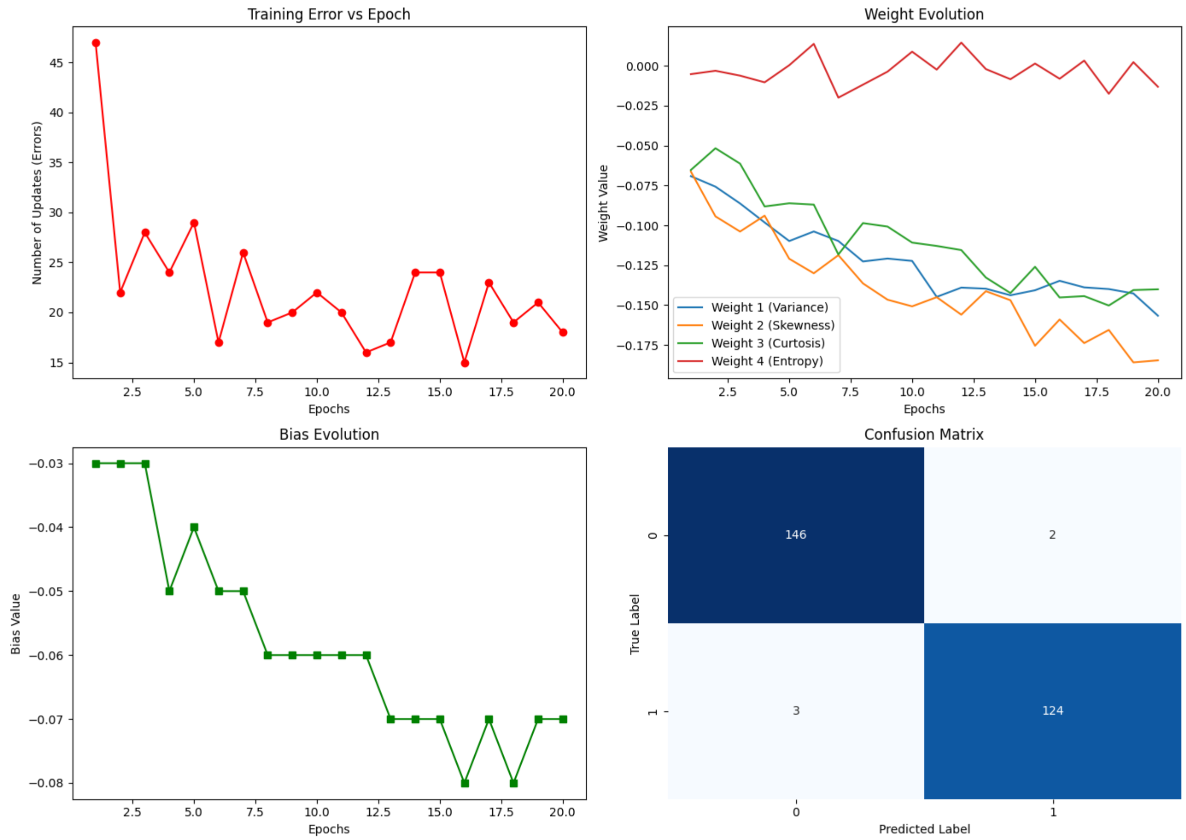
According to the boxplots, most outliers belong to Curtosis, which has several values that go as high as 10, while Variance has the least scattered and more symmetric data distribution out of the four parameters. Skewness has the widest interquartile range, but without any extreme outliers. Finally, Entropy is more grouped than its competitors, as it has a couple of outliers below −6.

Additional Pairwise Scatter Plots



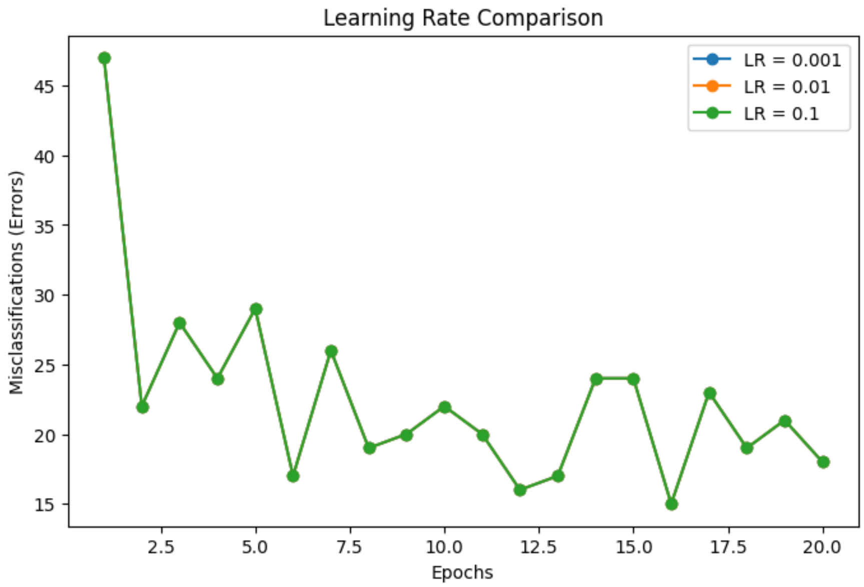
In the comparison between variance and kurtosis as well as skewness and kurtosis, the strongest discrimination is seen as the two classes show distinct clustering in the diagonal clusters. However, even when comparing variance to entropy, the classes can also be seen to be fairly well separated along the diagonal. In the case of skewness contrasted to entropy and kurtosis compared to entropy, there seems to be a higher degree of overlap between the classes as entropy alone does not represent a good discriminative criterion, which is consistent with its low correlation with class in the correlation table.

Training Error vs Epoch, Weight Evolution, Bias Evolution and Confusion Matrix



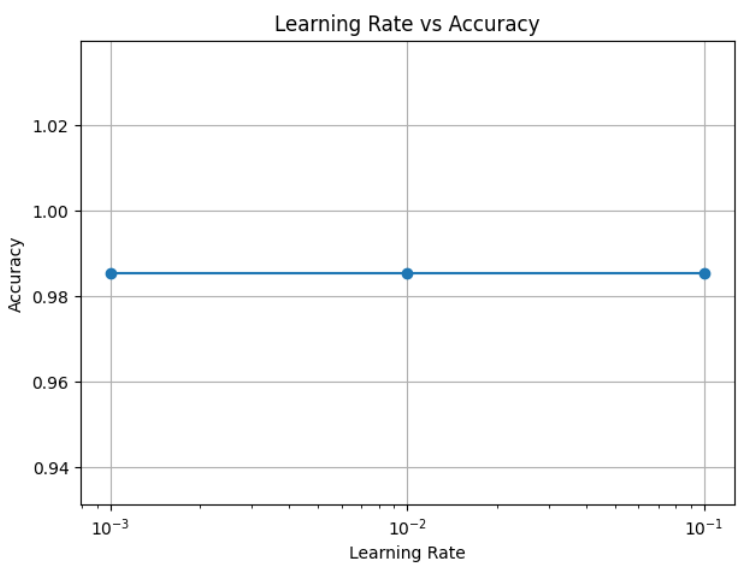
The graph depicting the Training Error versus Epoch displays a rapid decline in the number of misclassifications from 47 to 22 during the first two epochs, followed by fluctuating error values between approximately 15 and 29; this demonstrates that the classes are not perfectly linearly separable, as they do not completely converge to zero. The Weight Evolution graph indicates that the weights for Variance and Skewness decrease steadily and stabilize in the range of -0.15 and -0.18, while the Bias Evolution graph shows that bias decreases in small steps in the process of training, dropping from -0.03 to -0.07. The Confusion Matrix shows a presence of only 2 false positives and 3 false negatives out of 275 test samples, thus confirming a high classification quality indicated in the assessment metrics.

Learning Rate Comparison



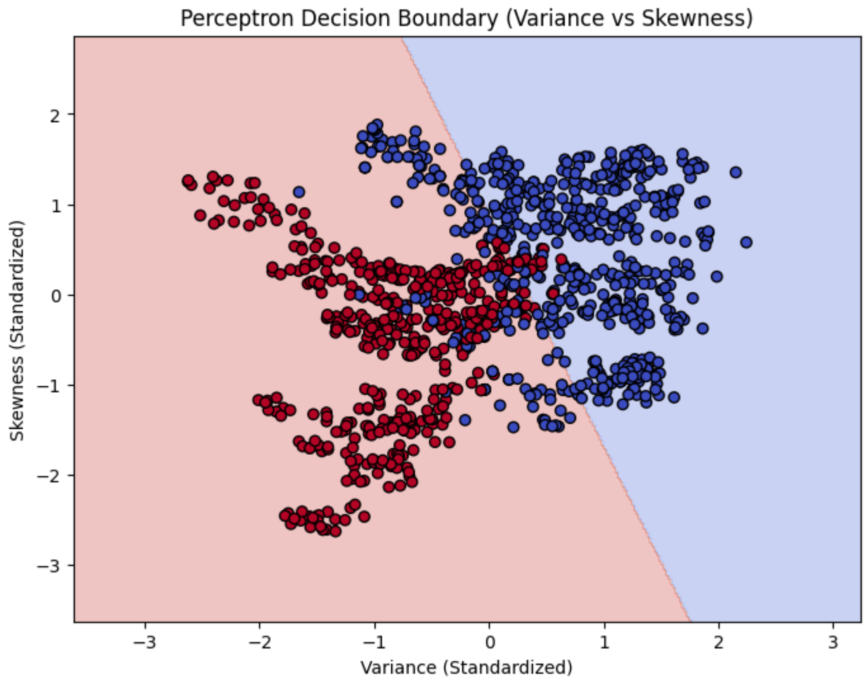
The misclassification curves resulting from learning rates 0.001, 0.01, and 0.1 are almost the same. This shows that, with zero-initialized weights, each weight update has the same sign, which means that predictions at each step are the same as well. This indicates that, for this data set, the convergence behavior of the perceptron is independent of the learning rate and only depends on the magnitude of the weights at the last instance.

Learning Rate vs Final Accuracy



The final test accuracy fluctuates around the value of 0.985 for all three of the used learning rates (0.001, 0.01, 0.1) in a manner that corroborates the earlier conclusion that only the update magnitude, but not the update direction, depends on . Therefore, the choice of learning rate has impact on the time the weights reach their final values which does not influence classifier performance.

Decision Boundary (Optional)



The perceptron uses variance and skewness to learn a linear decision line that distinguishes real authentic banknotes (colored blue) from counterfeit banknotes (colored red). This method confirms the basic principle that higher variance means the banknote is real. The band of misclassified points around the decision line also indicates that these two features are not completely linearly separable. Thus the training error seen in the previous plot does not reach zero in this case.

7. Performance Tables

Training Summary

Dataset Size	1372 samples
Train/Test Split	1097 / 275 (80% / 20%)
Learning Rate	0.01
Epochs	20
Final Weights	[-0.1567, -0.1846, -0.1401, -0.0132]
Final Bias	-0.0700
Accuracy	0.9818
Precision	0.9841
Recall	0.9764
F1-score	0.9802

Epoch-wise Learning (Variance and Skewness weights shown; $\eta = 0.01$)

Epoch	Errors	Weight 1 (Variance)	Weight 2 (Skewness)	Bias
1	47	-0.0692	-0.0663	-0.0300
2	22	-0.0758	-0.0944	-0.0300
3	28	-0.0863	-0.1039	-0.0300
4	24	-0.0982	-0.0939	-0.0500
5	29	-0.1098	-0.1210	-0.0400
⋮	⋮	⋮	⋮	⋮
16	15	-0.1348	-0.1590	-0.0800
17	23	-0.1389	-0.1738	-0.0700
18	19	-0.1399	-0.1656	-0.0800
19	21	-0.1427	-0.1859	-0.0700
20	18	-0.1567	-0.1846	-0.0700

8. Discussion and Conclusion

The Single Layer Perceptron developed from the start on the Banknote Authentication data attained a high accuracy of 98.18 percent in testing, with precision, recall, and F1-score metrics being 0.97 or higher, confirming that the four characteristics that comprise the data set (variance, skewness, curtosis, and entropy) are almost, though not entirely, linearly separable. In this model, training error does not reach zero value within 20 epochs, demonstrating 15-29 errors per epoch after the first initial steep performance peak, which indicates that some values were placed on the negative side of the separation line. The learning rate comparison illustrated that for the presented perceptron content, the quantity of the misclassifications did not differ from zero for the learning rates 0.001, 0.01, and 0.1. In this case, the learning rate influences only the value of the updates and not their sign. In case only two factors (Variance and Skewness) are considered for visualization of the boundary, results of the model show acceptable performance but including two additional features creates more clear borders. The total result of the custom built application (98.18) is very close to that of the built-in Scikit-learn perceptron (98.55).

9. References

1. F. Rosenblatt, "The Perceptron," *Psychological Review*, 1958.
2. I. Goodfellow, Y. Bengio and A. Courville, *Deep Learning*, MIT Press, 2016.
3. C. M. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006.
4. S. Haykin, *Neural Networks and Learning Machines*, Pearson, 2009.
5. UCI Machine Learning Repository – Banknote Authentication Dataset.
6. Scikit-learn Documentation: Perceptron.